

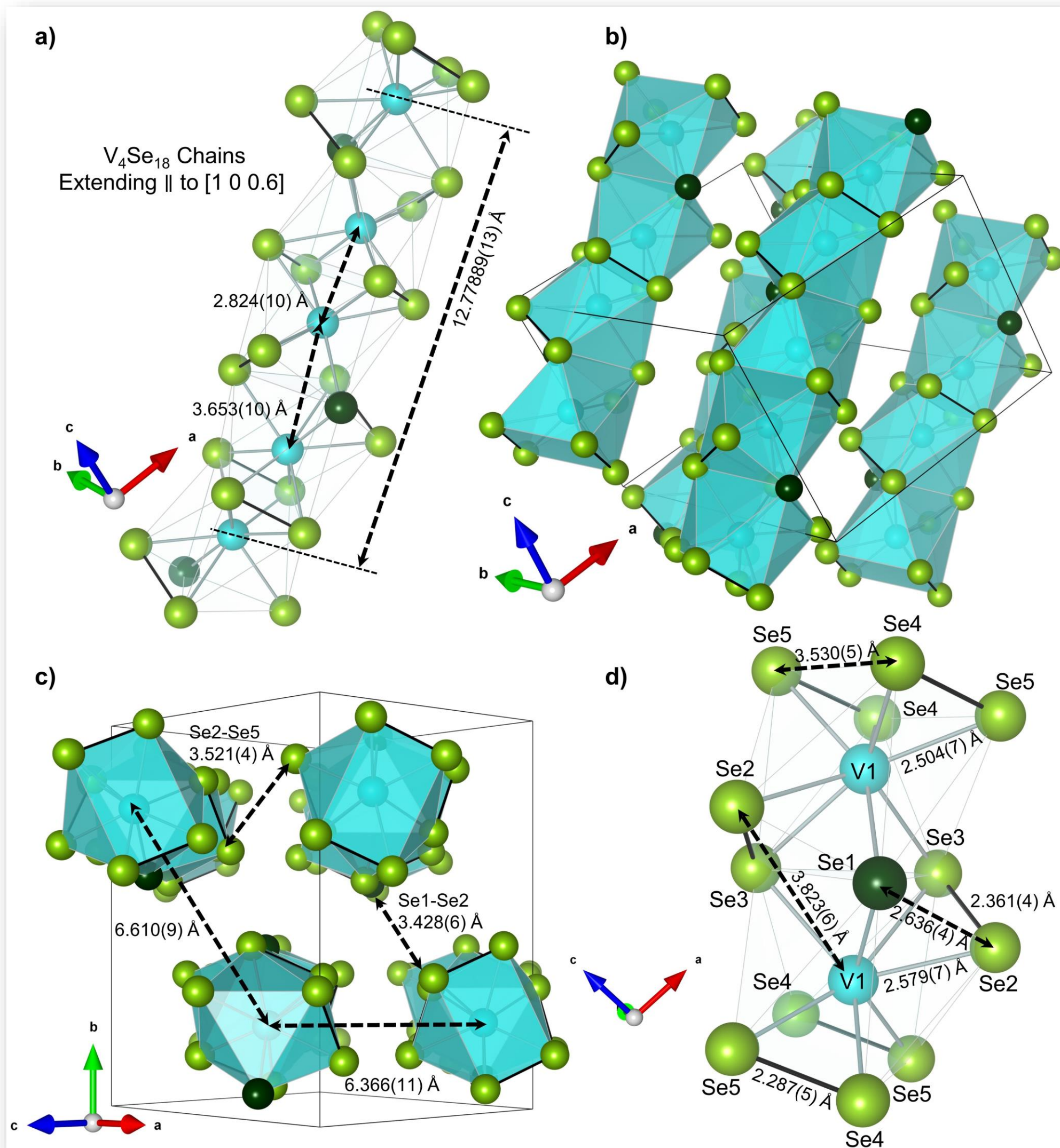
Fast Mg^{2+} Insertion Kinetics in V_2Se_9

Structural Evidence for Rapid & Reversible Inter-Chain Mg^{2+} Incorporation

MA Wright, AV Chadwick, JB Claridge, L Hardwick, MJ Rosseinsky* & coworkers, *J Mater. Chem. A*, 2024, **12**, 32349-32358
Materials Innovation Factory & Stephenson Institute for Renewable Energy, University of Liverpool, L69 7ZD, United Kingdom

Magnesium batteries promise higher volumetric capacity than Li^+ and Na^+ , while using low-cost, earth-abundant materials. So far, commercialization is limited by sluggish Mg^{2+} insertion kinetics at the cathode: the divalent ion's high charge-per-size density strongly interacts with the anion sub-lattice of the host material, slowing diffusion and hindering reversible insertion. This motivates the design open frameworks with polarizable anionic sub-lattices.

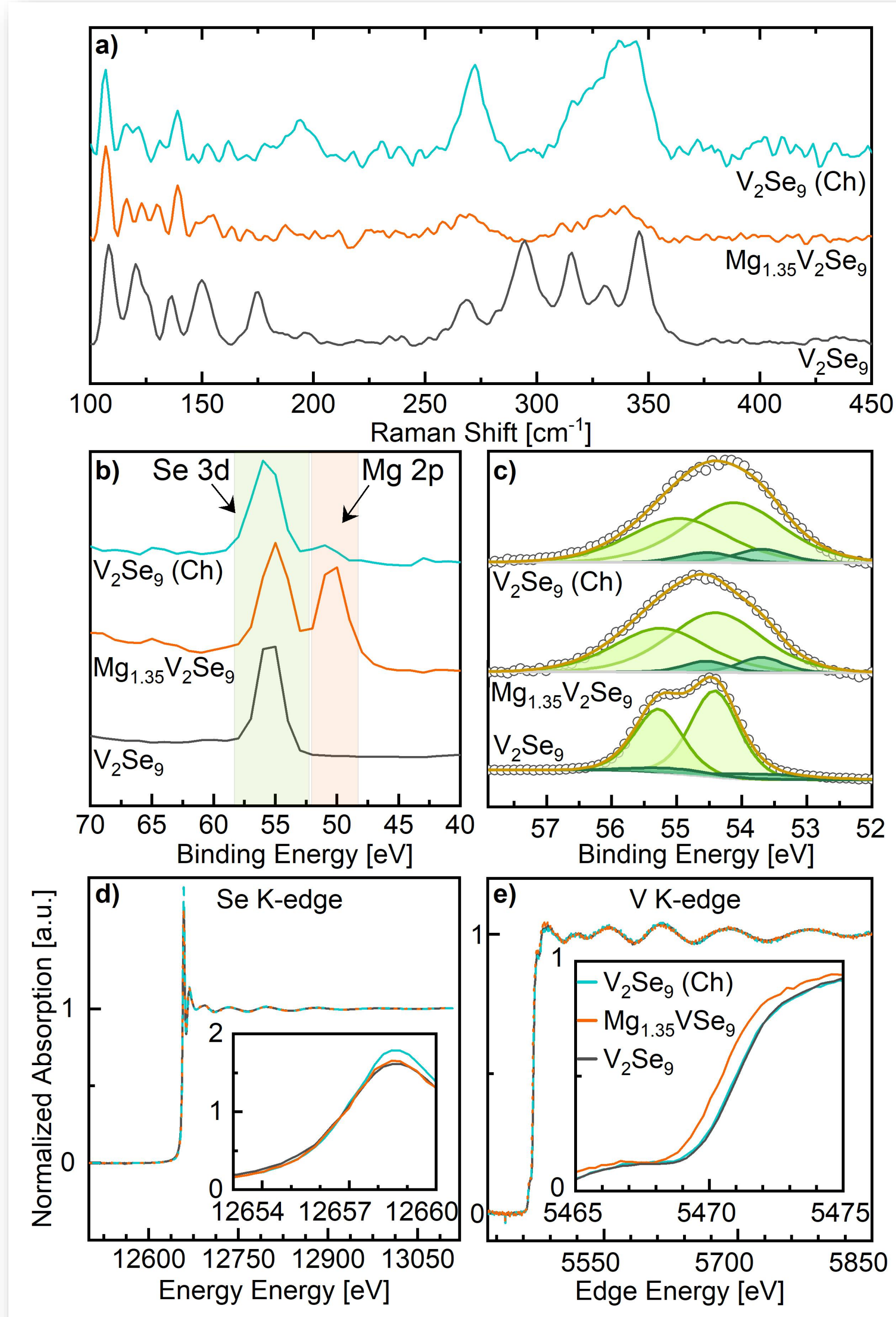
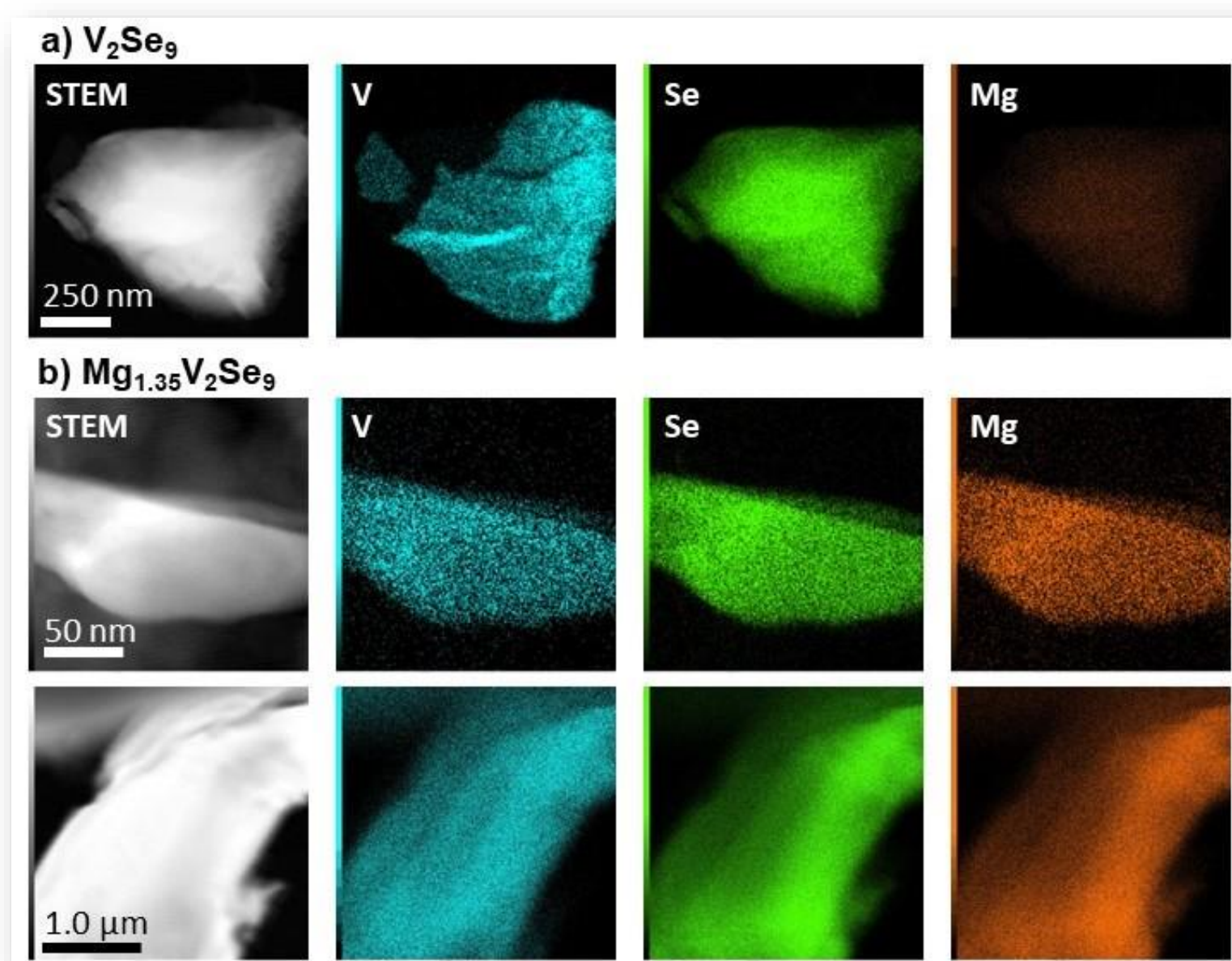
V_2Se_9 is employed as a quasi-one-dimensional electrode material capable of reversible insertion of up to $\sim 1.6 \text{ Mg}^{2+}$ per formula unit with unusually fast diffusion coefficients ($\approx 10^{-10}$ – $10^{-12} \text{ cm}^2 \text{ s}^{-1}$), demonstrating that highly polarizable selenide frameworks can overcome the kinetic limits that usually hamper multivalent batteries.



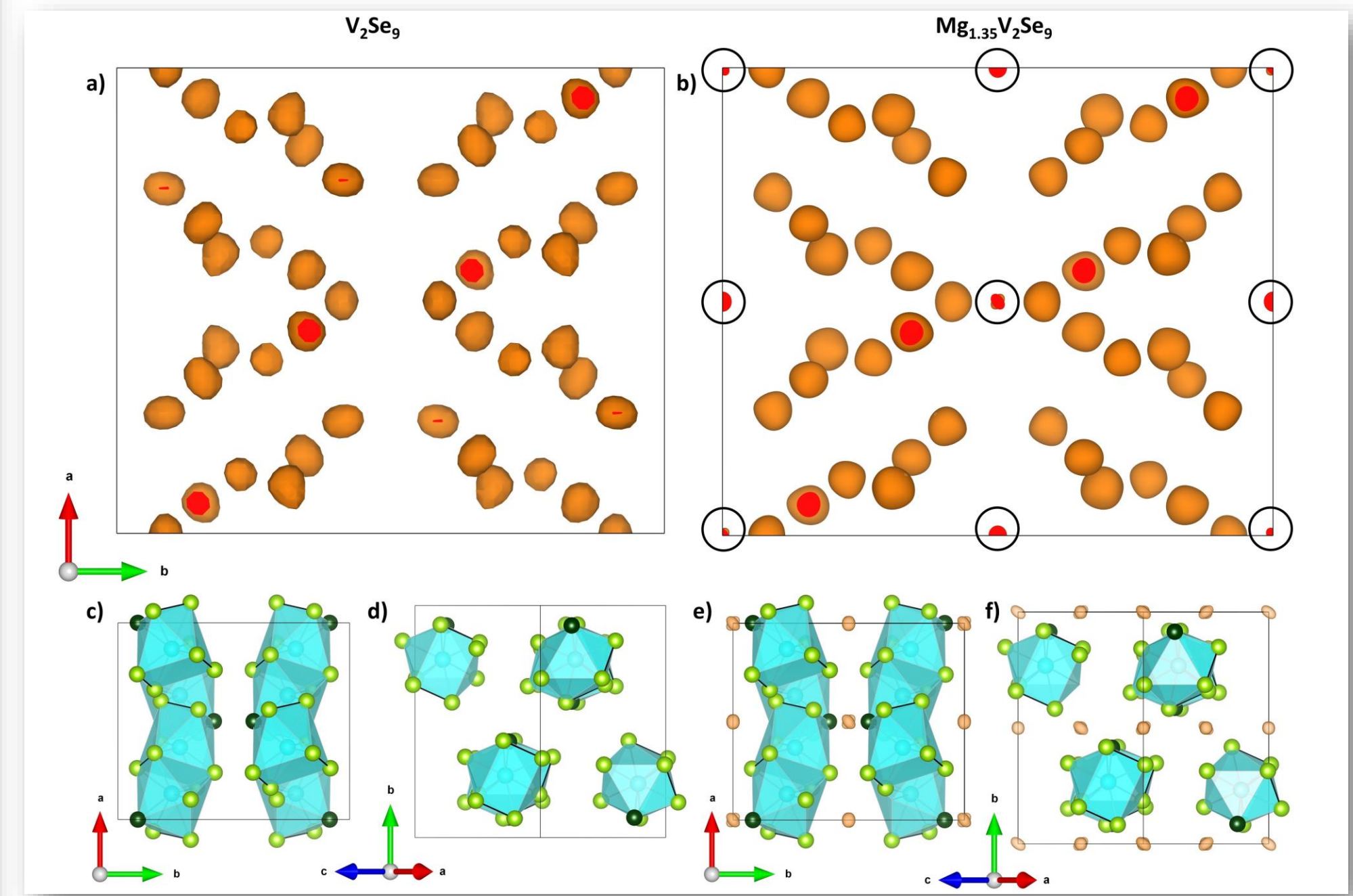
Electrochemistry: 0.8–2.2 V vs Mg^{2+}/Mg ; galvanostatic cycling and PITT to extract Mg diffusion coefficients

Synchrotron X-ray diffraction: Rietveld & Maximum Entropy Method to locate Mg sites

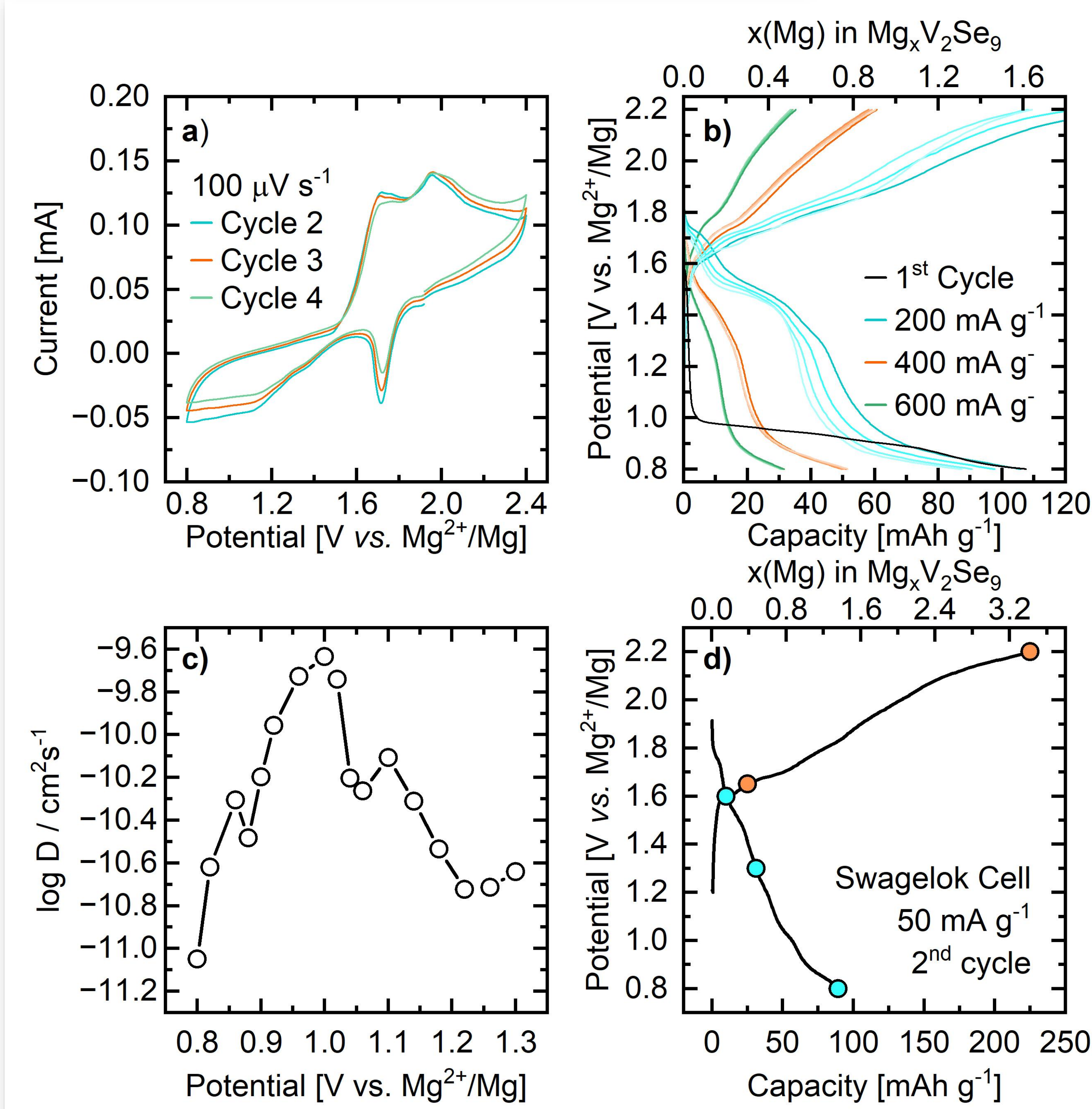
Spectroscopy & microscopy: STEM-EDX, ICP-OES, XPS and XANES to confirm composition, Mg location, and reversible V redox.



Redox mechanism elucidated by XANES and XPS, which shows Se^{2-} (Se_2) $^{2-}$ remain essentially redox-inactive over the tested window, while V^{5+} is reduced on discharge and oxidized on charge, indicating reversible cationic redox

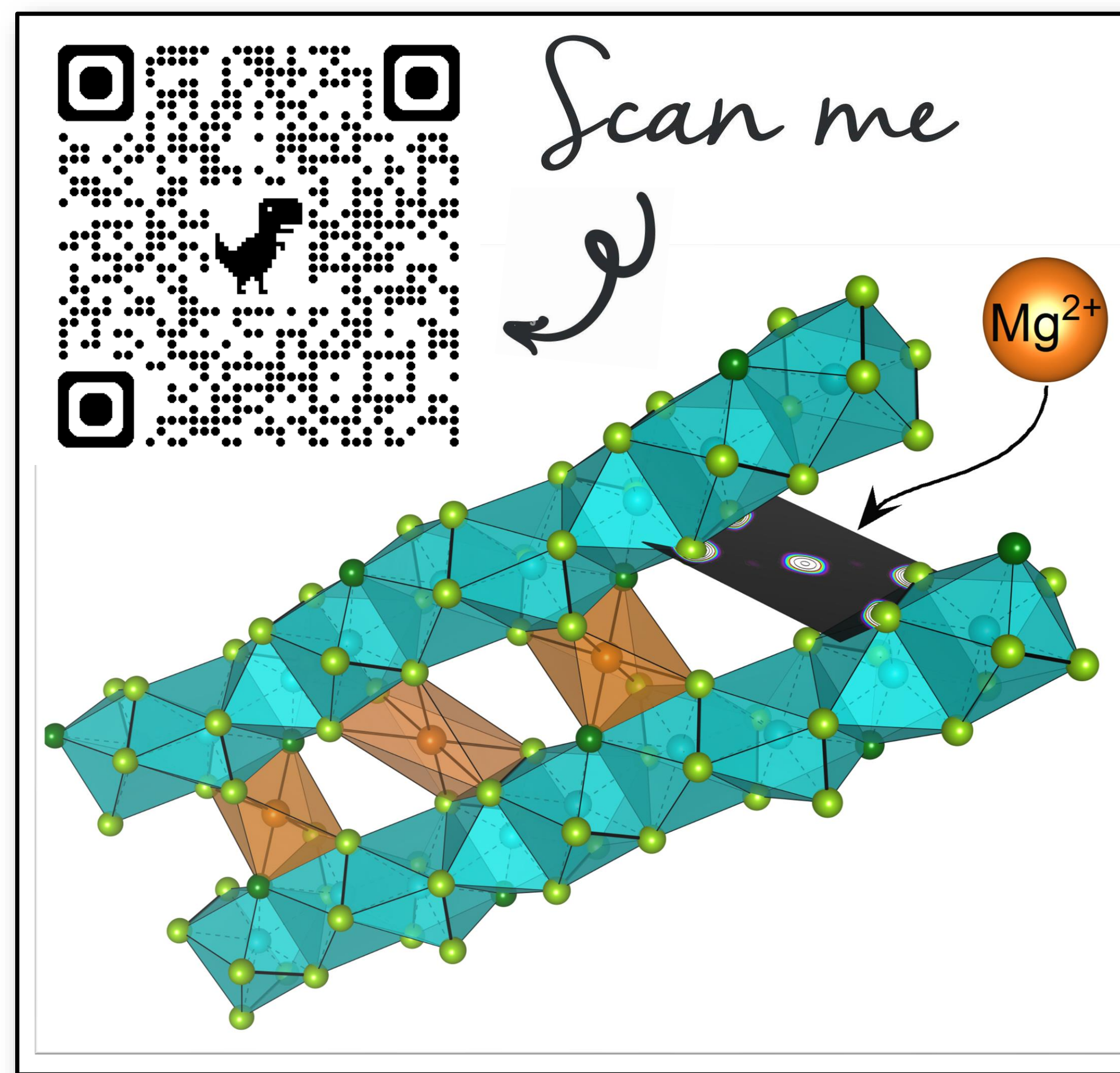


Insertion sites and structure retention: Two distinct octahedral Mg sites (Mg1 , Mg2) identified from electron density distribution maps using Maximum Entropy Method calculations. These sites occupy vdW gaps between $[\text{V}_4\text{Se}_{18}]_n$ chains; C2/c symmetry and the V-Se framework are preserved on (de)insertion, indicating an insertion-dominated mechanism.

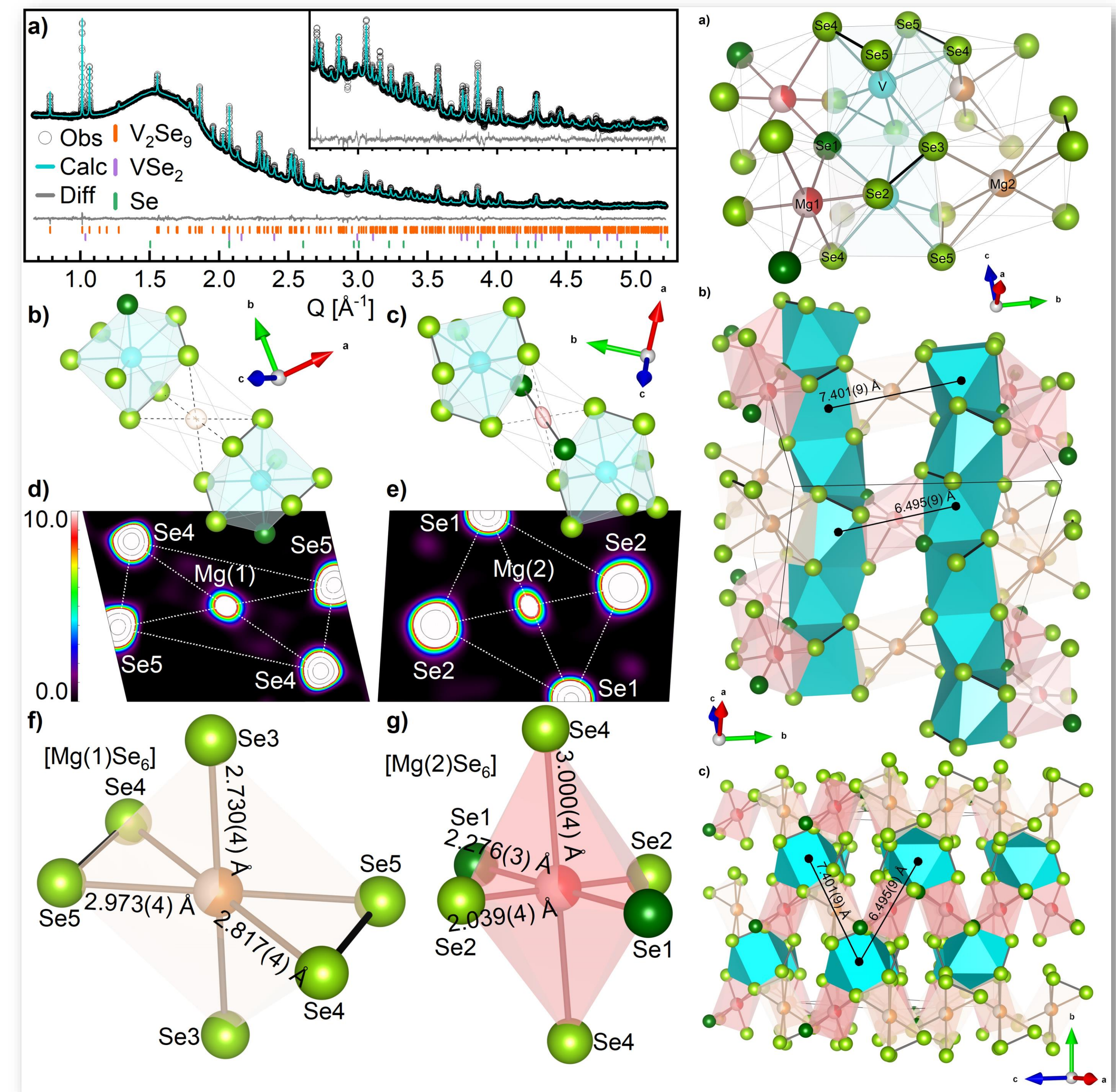


Capacity and stoichiometry: Reversible insertion ≈ 1.3 – 1.6 Mg^{2+} per V_2Se_9 (≈ 90 – 107 mA h g^{-1}) achieved by 0.8 V, with bulk composition consistent with $\text{Mg}_{1.20(3)}\text{V}_2\text{Se}_{9.4(5)}$ confirmed by ICP-OES
Fast kinetics: PITT-derived Mg^{2+} diffusion coefficients between 10^{-10} – $10^{-12} \text{ cm}^2 \text{ s}^{-1}$, faster than benchmark multivalent oxide & sulfide cathodes (e.g., Chevrel phase Mo_6S_8 , VS_4 , and V_2PS_{10}), and comparable to monovalent Li^+/Na^+ insertion hosts

Takeaway: V_2Se_9 enables fast, reversible Mg^{2+} insertion by combining large vdW gaps, low-charge-density (Se_2) $^{2-}$ and Se^{2-} coordination, low migration barriers, and reversible V cation redox, demonstrating that tailored anion chemistry and low-dimensional architectures can overcome the kinetic bottleneck associated with multivalent cathodes. V_2Se_9 demonstrates that replacing $\text{O}^{2-}/\text{S}^{2-}/\text{Se}^{2-}$ with more polarizable anions (Se_2) $^{2-}$ and can substantially improve multivalent ion kinetics. This results motivates targeted exploration of low-dimensional selenides and mixed-chalcogen frameworks for multivalent energy storage.



Minimal local distortion: MgSe_6 octahedra impose only small changes to average V-Se and Se-Se bond lengths; unit cell volume changes are $< 1\%$, explaining the low mechanical strain during cycling.



Diffusion pathways: Bond-valence pathway analysis maps interconnected 1D–3D diffusion channels through the large vdW spaces; calculated local barriers span ≈ 0.14 – 0.74 eV , consistent with the experimentally fast diffusion rates.

Anion role: The highly polarizable $\text{Se}^{2-}/(\text{Se}_2)^{2-}$ framework reduces strong electrostatic trapping of Mg^{2+} , increasing channel sizes and lowering migration barriers relative to sulfides/oxides.

Site filling sequence: Larger, lower-charge-density-coordinated Mg1 is occupied earlier in discharge, then Mg2 , matching capacity-dependent kinetics observed in PITT.